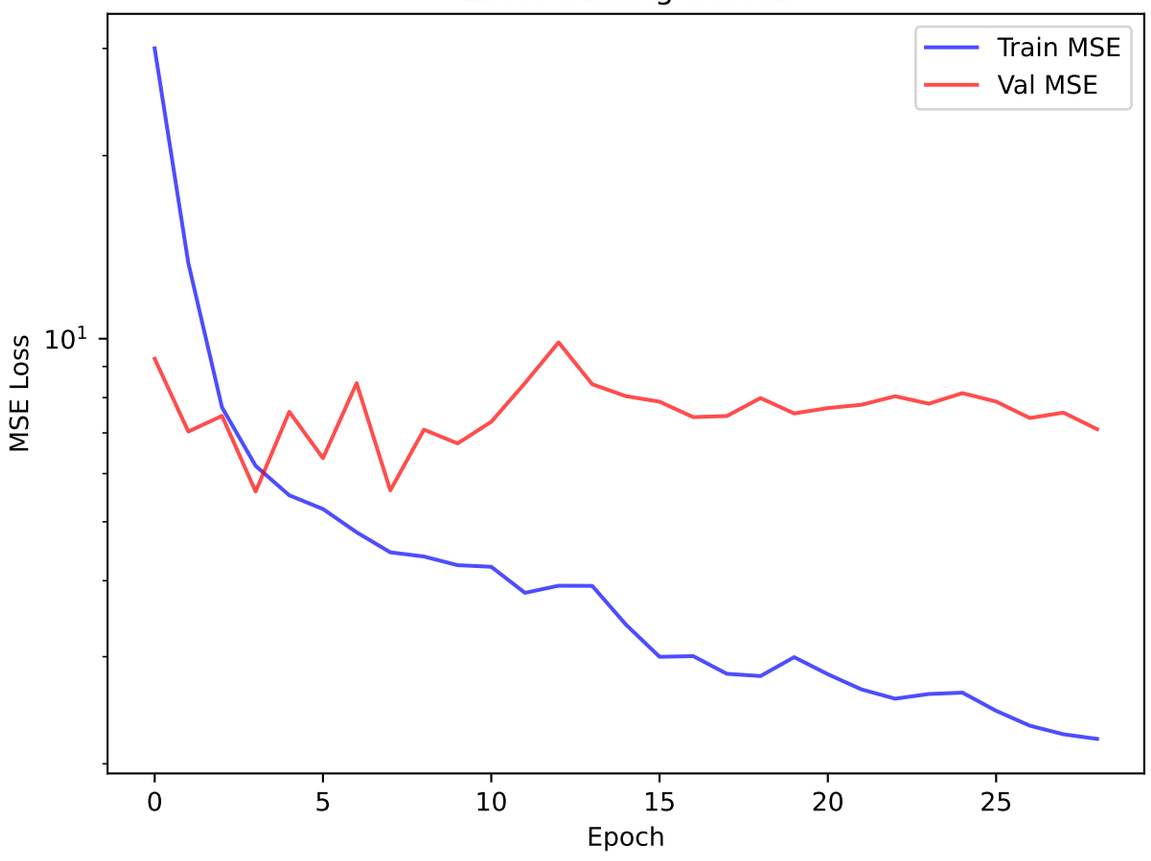
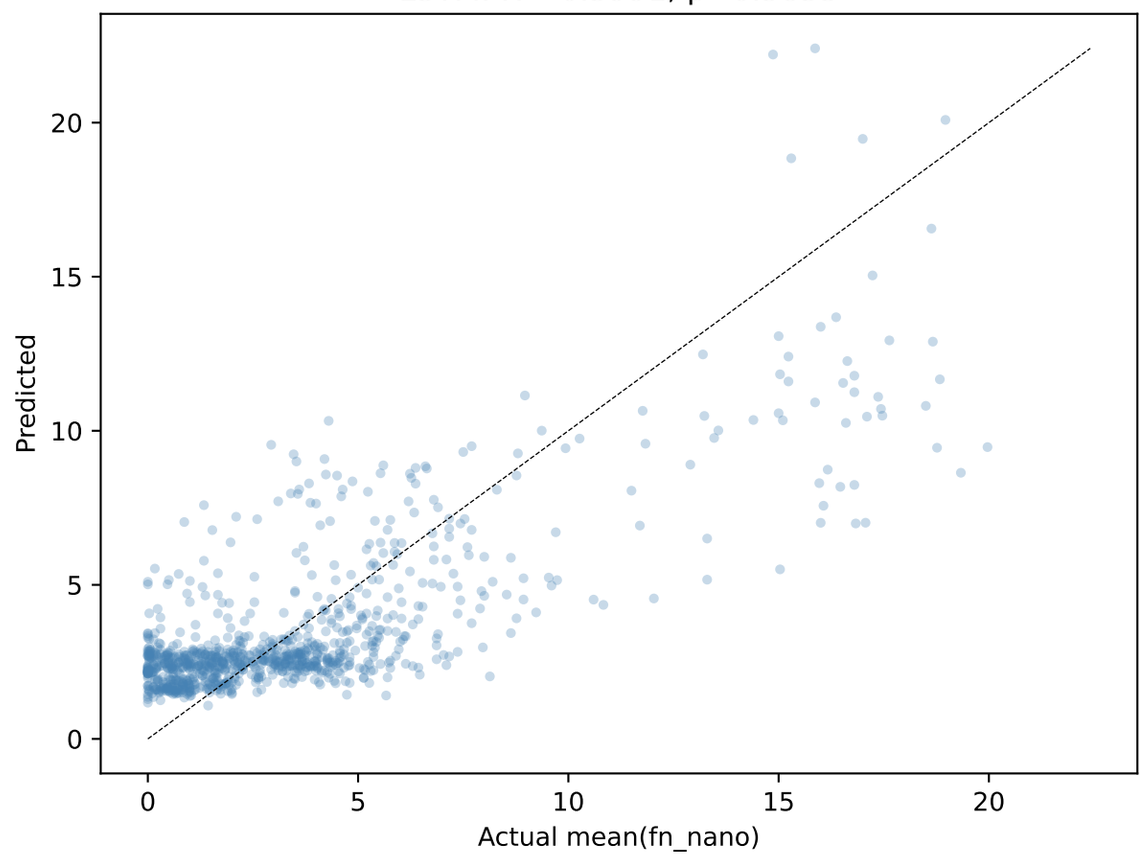
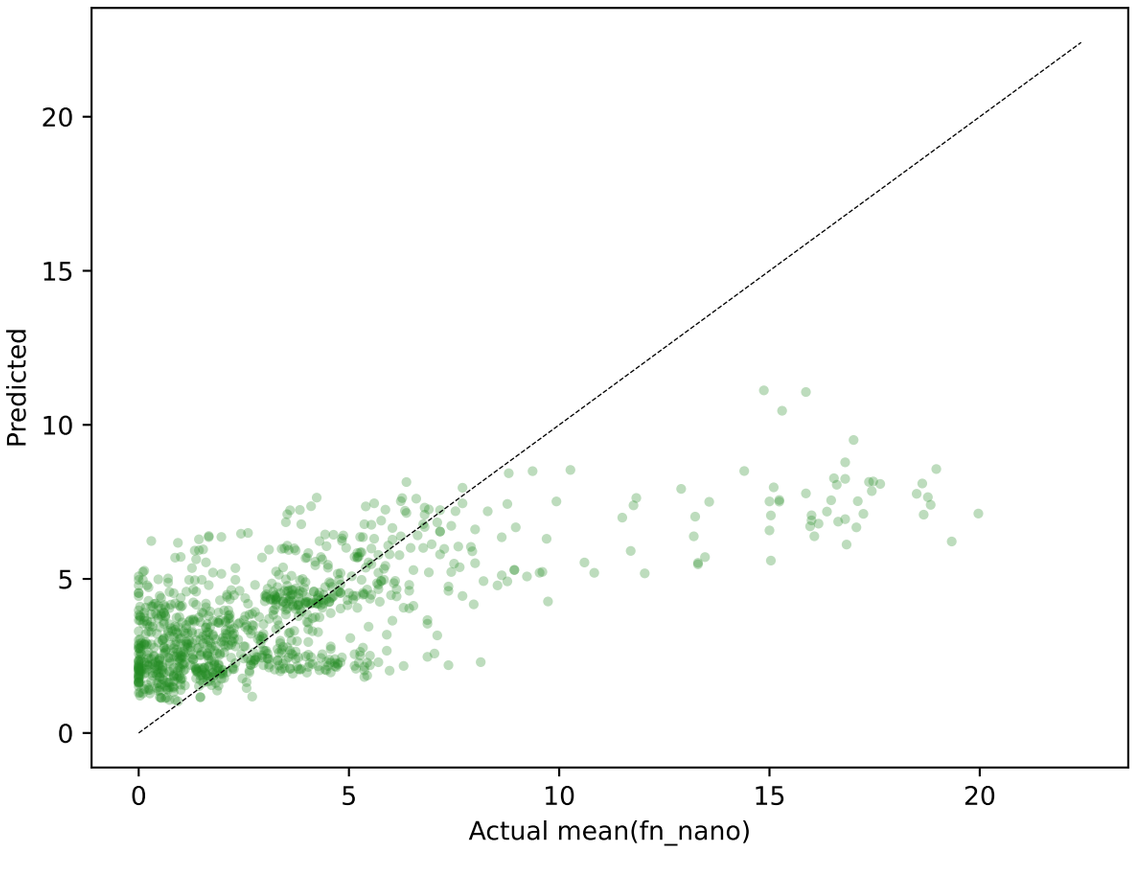
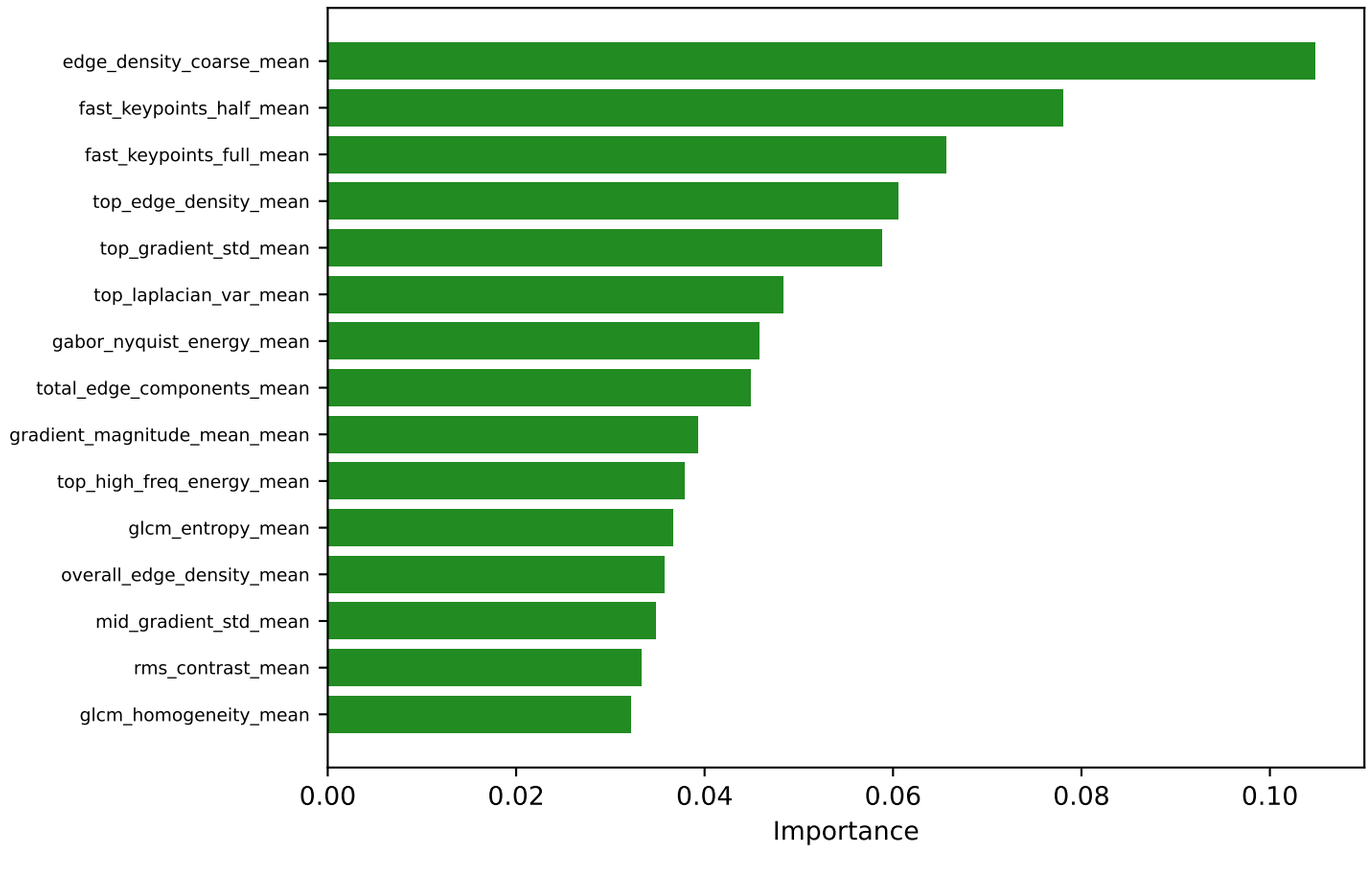
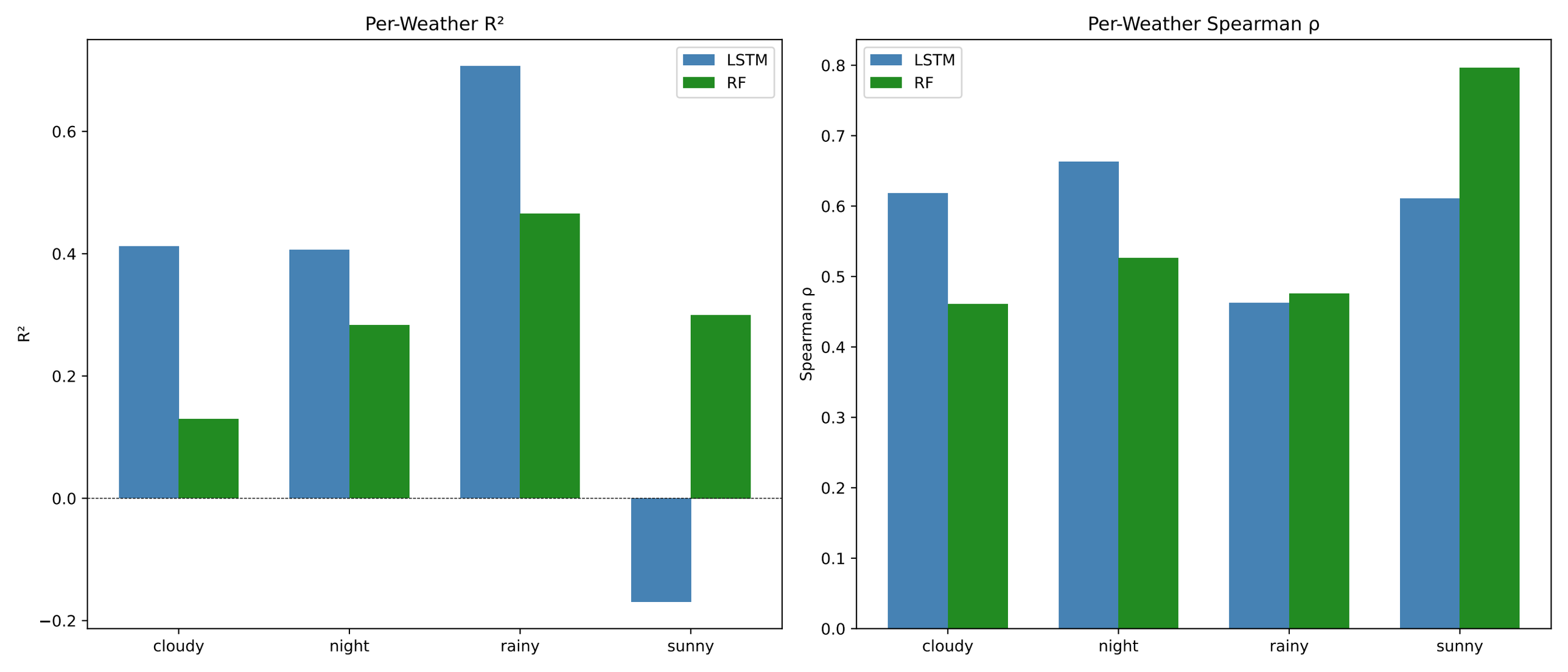

LSTM Training Curves
Train MSE
Val MSE
MSE Loss
10^1
0
5
10
15
20
25
Epoch


LSTM: R²=0.5991, ρ=0.5853
Predicted
0
5
10
15
20
Actual mean(fn_nano)


RF: R²=0.4314, ρ=0.5921
Predicted
0
5
10
15
20
Actual mean(fn_nano)


RF Top 15 Feature Importances
edge_density_coarse_mean
fast_keypoints_half_mean
fast_keypoints_full_mean
top_edge_density_mean
top_gradient_std_mean
top_laplacian_var_mean
gabor_nyquist_energy_mean
total_edge_components_mean
gradient_magnitude_mean_mean
top_high_freq_energy_mean
glcm_entropy_mean
overall_edge_density_mean
mid_gradient_std_mean
rms_contrast_mean
glcm_homogeneity_mean
0.00
0.02
0.04
0.06
0.08
0.10
Importance


Per-Weather R²
LSTM
RF
R²
0.6
0.4
0.2
0.0
−0.2
cloudy
night
rainy
sunny
Per-Weather Spearman ρ
LSTM
RF
Spearman ρ
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
cloudy
night
rainy
sunny

Per-Sequence R² (val set, non-overlapping windows)

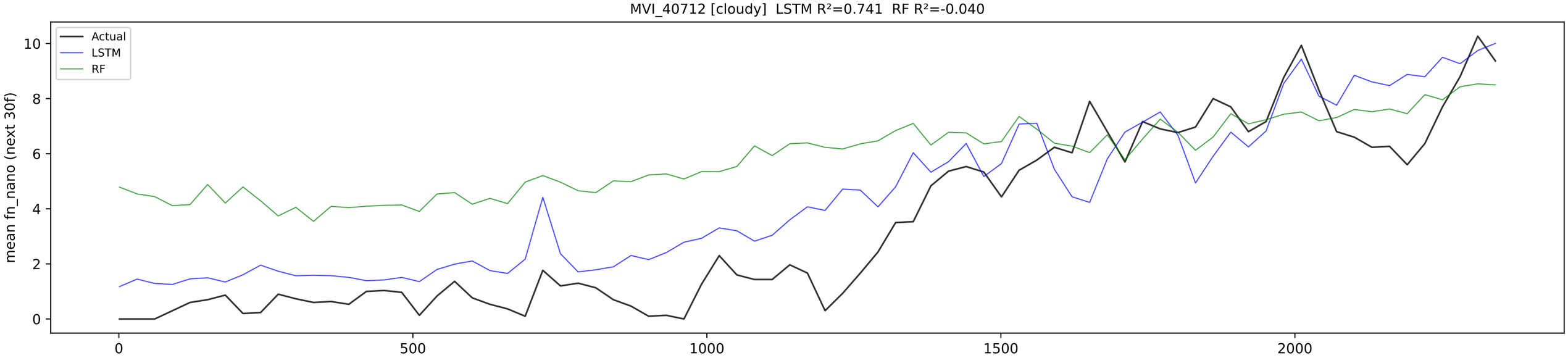
LSTM
RF

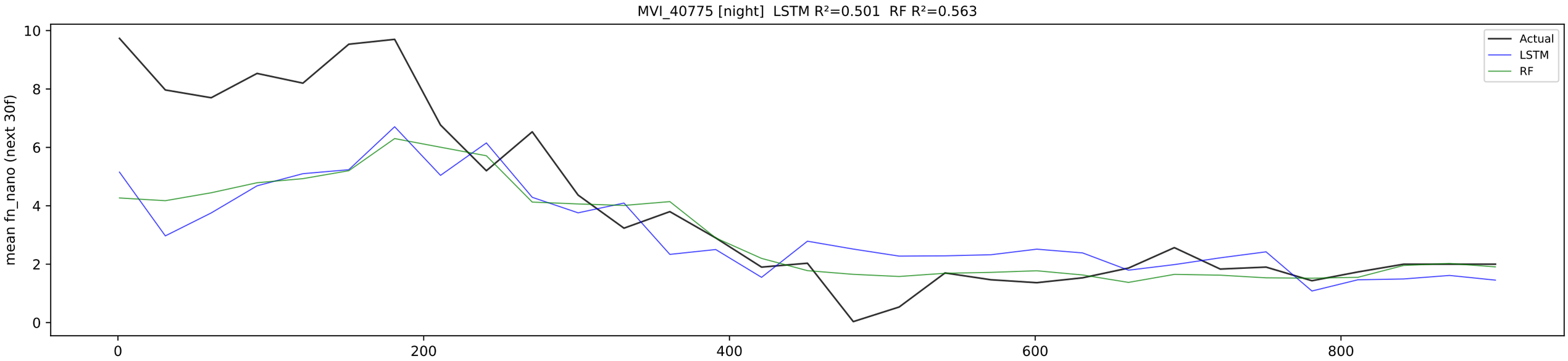
MVI_40712 [cloudy]
MVI_40775 [night]
MVI_40732 [night]
MVI_40191 [cloudy]
MVI_63553 [rainy]
MVI_40243 [cloudy]
MVI_63521 [rainy]
MVI_40992 [night]
MVI_20011 [sunny]
MVI_40244 [cloudy]
MVI_20012 [sunny]
MVI_40892 [rainy]
MVI_20032 [sunny]
MVI_40752 [night]
MVI_40172 [cloudy]
MVI_39361 [sunny]
MVI_39811 [night]
MVI_20051 [sunny]
MVI_63544 [rainy]
MVI_63525 [rainy]

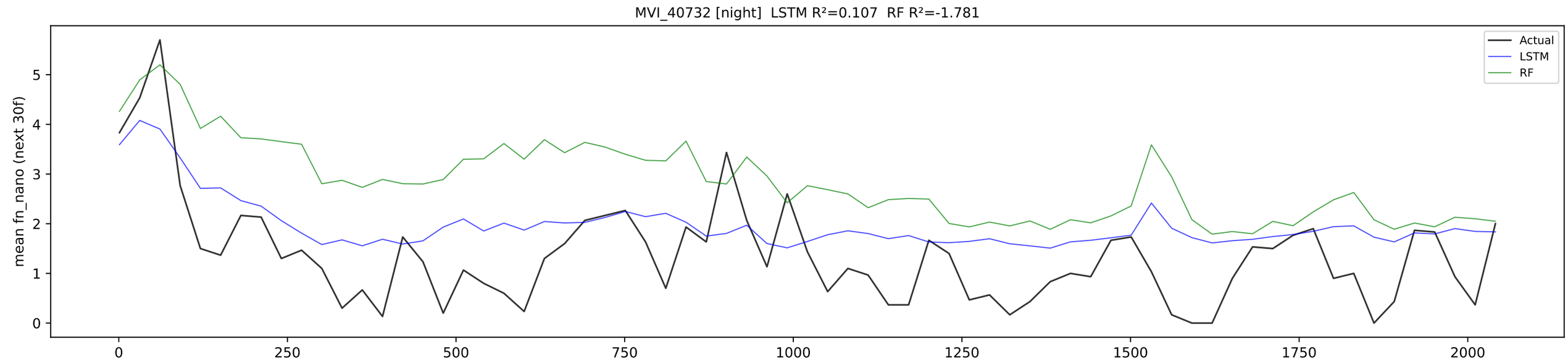
−40
−30
−20
−10
0

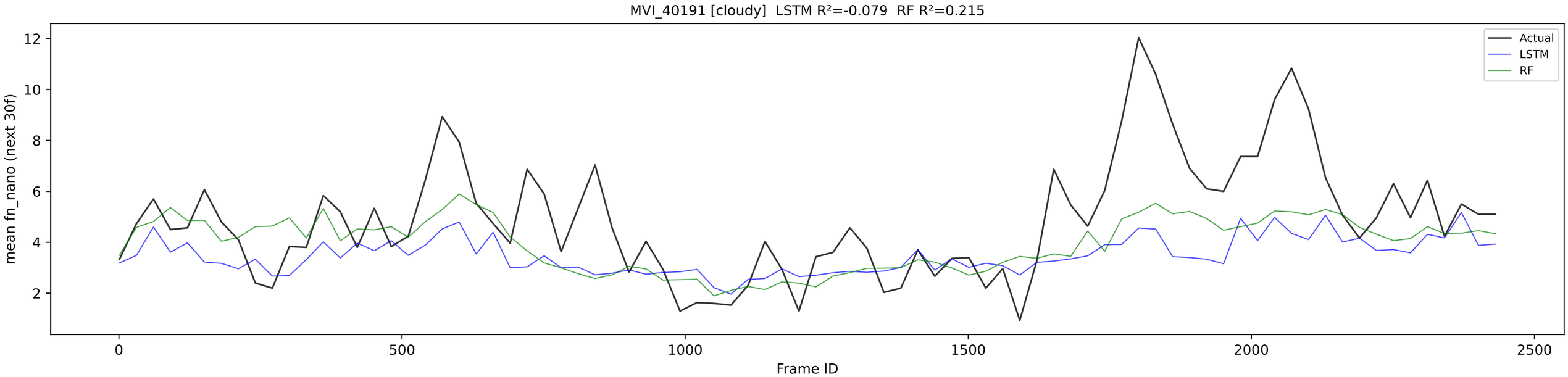
R²

Best Val Sequences — Actual vs Predicted









LSTM vs Random Forest — Shifted-Label Prediction

	R^2	MAE	Spearman ρ
LSTM (2-layer, h=64)	0.5991	1.753	0.5853
Random Forest (500 trees)	0.4314	1.988	0.5921
Baseline (predict mean)	-0.1189	3.094	0.0000

Task: predict mean(fn_nano) over next 30 frames from 30 frames of 20 features
Train stride: 15, Eval stride: 30 (non-overlapping)
Split: 80+ train / 20 val sequences (stratified by weather)
Normalization: StandardScaler fit on train only
No num_gt filter (preserves temporal continuity)
No per-sequence mean removal (avoids look-ahead bias)